TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Standard → slope-intercept: subtract the x-term, then divide every term by the number with y.
Point-slope → slope-intercept: distribute, then add or subtract to get y alone.
Intercepts: x-intercept: put 0 in for y. y-intercept: put 0 in for x.
Solve a formula for a letter: undo everything around that letter, doing the same to both sides.

1. Which equation is $4x + 2y = 10$ written in slope-intercept form?

Tip: get [[y]] alone: subtract, then divide every term

A. $y = 2x + 5$

B. $y = -2x + 5$

C. $y = -4x + 10$

Subtract $4x$: $2y = -4x + 10$. Divide by 2: $y = -2x + 5$.

2. What value of x makes $3x - 7 = 11$ true?

Tip: undo − with +, undo × with ÷

F. 4/3

G. 18

H. 6

Add 7: $3x = 18$. Divide by 3: $x = 6$.

3. $P = 2\ell + 2w$. What is ℓ when $P = 30$ and $w = 5$?

> **Tip:** put in the numbers you know, then solve

A. 20

B. 12.5

C. **10**

30 = 2ℓ + 10, so 2ℓ = 20 and ℓ = 10.

4. $C = 5/9(F - 32)$. What is C when $F = 50$?

> **Tip:** parentheses first, then multiply

F. **10**

G. 18

H. 50

50 − 32 = 18, and 5/9 · 18 = 10.